ARAVINDHAN S

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience developing and integrating Machine Learning, Deep Learning, and Generative AI solutions using Python. Experienced in building AI-powered applications, backend APIs, RAG pipelines, agentic workflows, data preprocessing pipelines, and asynchronous AI systems. Strong foundation in machine learning algorithms, model development from scratch, PyTorch, Scikit-Learn, FastAPI, SQL, and AI/LLM integration. Experienced in deploying, testing, monitoring, and troubleshooting AI applications and interested in solving real-world problems through practical AI solutions.

TECHNICAL SKILLS

Programming & AI Development

Python | Object-Oriented Programming (OOP) | AsyncIO | FastAPI | REST APIs | End-to-End System Design | AI-Generated Code Cleanup & Refactoring | Compute & Resource Estimation for Process Allocation

Machine Learning & Deep Learning

PyTorch | Scikit-Learn | Neural Networks | Machine Learning Algorithms | Deep Learning | Model Development from Scratch | Model Evaluation | Feature Engineering | Data Preprocessing | Computer Vision | YOLO | Vision Transformers (ViT) | Natural Language Processing (NLP) | Recommender Systems | Graph Neural Networks | GraphSAGE | KNN | Locality Sensitive Hashing (LSH)

Generative AI

Large Language Models (LLMs) | Transformers | Retrieval-Augmented Generation (RAG) | RAG Evaluation | Prompt Engineering | Agentic AI Systems | LangChain | LangGraph | CrewAI | Model Context Protocol (MCP) | Fine-Tuning | Vector Databases | FAISS | Chroma | General Vectorization

AI Infrastructure & Deployment

Docker | Celery | Redis | Redis Pub/Sub | Distributed Task Queues | Event-Driven Architecture | Asynchronous Inference | Git | GitHub Actions | Pytest | Prometheus | Input & Output Guardrails | Loop Engineering | Sandboxing | API Prefix Caching | Batch API Processing | Semantic RAG Caching

Data, Analytics & Databases

Data Cleaning | Data Preprocessing | Exploratory Data Analysis (EDA) | Missing Value Handling | Outlier Detection | Feature Scaling | Cross-Validation | PostgreSQL | MySQL | SQLAlchemy | Alembic

WORK EXPERIENCE

AI Systems Engineer

Vankaal

02/2026 - 08/2026

- Developed a real-time AI voice consultation system using Python, FastAPI, and Twilio for automated technical support.
- Integrated LLM-based communication capabilities with custom deterministic planning and business-logic components to maintain controlled AI decision-making.
- Built asynchronous backend workflows for handling real-time voice processing, conversation state, context tracking, and AI responses.
- Developed high-throughput WhatsApp automation workflows using the Meta Graph API and custom webhooks.
- Implemented backend state management and multi-tenant communication workflows for reliable processing of concurrent user requests.
- Integrated AI services into application workflows through APIs and asynchronous processing architectures.
- Intelligent Email Automation & Data Extraction: Architected an event-driven AI workflow that categorizes incoming emails, extracts structured information from PDFs and invoices, and generates context-aware responses using NLP pipelines.
- Built preprocessing workflows to transform unstructured documents and incoming data into structured information for downstream AI processing.
- Algorithmic Recommender Systems: Designed and implemented personalized recommendation systems using statistical machine learning, KNN, LSH, and graph-based similarity techniques.
- Developed recommendation algorithms from scratch using mathematical and machine learning fundamentals rather than relying entirely on high-level APIs.

Generative AI Developer

AI Futurix Innolabz

02/2025 - 02/2026

- Developed Python-based microservices for integrating and serving AI models through backend APIs.
- Built FastAPI services for AI-powered applications using asynchronous processing and REST-based API integration.

- Developed and integrated Generative AI solutions involving LLMs, RAG pipelines, agentic workflows, and AI infrastructure.
- Implemented data and message-processing workflows using AsyncIO, Redis, and asynchronous task architectures.
- Used Docker to package and maintain consistent AI application environments across development and deployment.
- Developed secure API endpoints using OAuth2 authentication and JWT-based authorization.
- Implemented automated testing using Pytest to validate AI application components and backend services.
- Monitored AI application and service performance using Prometheus and investigated application-level issues.
- Worked with AI models and backend systems to integrate machine learning capabilities into practical software applications.

KEY PROJECTS

Multichannel Communication Automation (WhatsApp & Email)

Tech Stack: Python, Webhooks, Message Queuing, Fast API, LLMs

- Built an automated event-driven system to handle incoming customer inquiries across WhatsApp and email, ensuring rapid and consistent responses.
- Implemented robust webhook integrations and background message queuing to reliably process high volumes of simultaneous incoming messages.
- Streamlined customer support workflows by classifying and routing requests automatically without human intervention.

Real-Time IVR (Interactive Voice Response) Automation

Tech Stack: Python, WebSockets, Local VAD, STT/TTS, RAG, Semantic Caching

- Architected a fully streaming automated telephone pipeline integrating local Voice Activity Detection (VAD), voice separation, STT, and TTS.
- Optimized API transcription costs by deploying local VAD to filter out silence, ensuring only active speech payloads are streamed to the STT model.
- Utilized WebSocket connections and queue management to process live audio streams with minimal latency.
- Optimized LLM context window limits by providing only recent interactions in the active prompt, building a running RAG system to dynamically retrieve relevant historical conversation context.
- Implemented a distributed global semantic cache across all active sessions to instantly resolve high-frequency queries (e.g., pricing), bypassing LLM inference to significantly reduce API costs.
- Maximized provider-side API prefix caching efficiency by strategically ordering prompt text to reduce inference latency and overall costs.
- Ensured stable and continuous voice interactions during peak call volumes by separating API processing from inference.

Voice-to-Voice AI Conversational System

Tech Stack: Python, Voice Processing, Conversational AI, Async Workflows

- Architected a seamless voice chat application allowing users to interact directly with an AI assistant in real-time.
- Optimized the processing pipeline to reduce response latency, creating a natural, human-like conversation flow.
- Managed conversation state and context tracking to ensure the AI maintained accurate and relevant responses throughout the dialogue.

Graph-Based Recommendation Engine

Tech Stack: Python, Graph Neural Networks, Text Embeddings

- Designed a personalized recommendation system that matches users with relevant items based on complex node relationships.
- Integrated text-based embeddings to better understand, vectorize, and categorize unstructured data for more accurate suggestions.
- Built the core recommendation logic and mathematical aggregations to ensure complete control over the matching and retrieval processes.

EDUCATION

Bachelor of Technology / Bachelor's Degree in Artificial Intelligence and Data Science

Dr. N.G.P. Institute of Technology

10/2021 - 04/2025

GPA: 8.0 / 10

CORE STRENGTHS

- Machine Learning & Deep Learning
- AI Application Development
- Data Preprocessing & Analysis
- RAG System Development
- Distributed AI Workflows
- Generative AI & LLM Integration
- Model Development from Scratch
- AI / ML API Integration
- AI Backend Engineering
- Model Evaluation & Experimentation

- Algorithmic Problem Solving
- Mathematical Foundations of AI